

BIBHU MISHRA

Principal Architect | Cloud, DevOps & Generative AI | Enterprise Delivery Leadership

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EXECUTIVE SUMMARY

Principal-level technology leader with 18+ years spanning hands-on engineering, enterprise solution architecture, and delivery leadership of onshore/offshore teams up to 50 — currently shipping production GenAI solutions (agentic RAG, fine-tuned LLMs, MCP-based tooling) at Microsoft. Proven record of turning ambiguous business goals into secure, scalable cloud-native and AI platforms, then guiding the teams that build, ship, and operate them. Trusted advisor to enterprise customers including Microsoft and Tenet Healthcare — owning roadmaps, managing expectations and escalations, and reporting to executive stakeholders. Deep domain grounding in Healthcare (EHR, HL7, HIPAA) and Content Management, paired with current expertise in Azure, Kubernetes, Terraform, and LLMOps. Equally effective defining architecture, mentoring engineers, and owning multi-million-dollar delivery commitments.

CORE COMPETENCIES

Leadership & Delivery: Team leadership & mentoring (teams up to 50) · Program & delivery management · Customer & stakeholder management · Pre-sales / RFP–RFI response · Governance & delivery excellence · Agile & Scrum

Architecture & Cloud: Enterprise & solution architecture · Azure (AKS, App Services, Functions, API Management, Front Door, Cosmos DB) · Kubernetes & containers · Terraform / IaC · Microservices

Generative AI & LLMOps: Agentic AI & RAG architecture · Open AI SDK / LangChain / LangGraph / AutoGen · LLM fine-tuning · MLflow / Langfuse observability · Azure OpenAI · Azure AI Search · Microsoft NLWeb · MCP (Model Context Protocol)

DevSecOps & Reliability: CI/CD (Azure DevOps, GitHub) SAST / DAST / SCA (SonarQube, CodeQL, Snyk) Zero Trust / Microsoft SFI · Prometheus, Grafana, Azure Monitor · SRE

Programming & Scripting: Strong hands-on experience with Node.js, Python, Vue.js, NestJS, FastAPI, FastMCP, and shell scripting, applied to building scalable applications, backend services, automation workflows, and CI/CD pipelines.

Domain Expertise: Healthcare (EHR, HL7, HIPAA, EMR) Content Management & Digital Marketing · Capital Markets · Utility (AMI / MDM) Credit (PLCC)

PROFESSIONAL EXPERIENCE

Microsoft (via LTI Mindtree)

Dallas, TX

Principal Azure Solution Architect — Microsoft Marketing Web (GDC)

Sep 2022 – Present

Azure solution architect and delivery owner for a portfolio of 50+ Microsoft marketing websites/services, defining target-state architecture and governing cloud modernization, platform engineering, application security, deployment, observability, and SRE. Accountable for translating business and operational requirements into secure, resilient, scalable Azure solutions that improve customer experience and engineering velocity.

- **Owned the Azure platform architecture and technical roadmap for Microsoft's GDC Web organization**, establishing reference patterns, nonfunctional requirements, and operational guardrails while mentoring engineers and governing availability, reliability, security, and performance across the marketing web estate.
- **Defined and delivered the target-state Azure architecture for migrating 40+ Microsoft.com blog and news sites to AKS, reducing infrastructure cost by 40%.** Designed reusable, self-healing, auto-scaling Kubernetes landing patterns with Terraform and Helm, with identity, authorization, workload security, and software-supply-chain controls implemented through Microsoft Entra ID authentication, RBAC, pod security policies, and container image scanning.
- **Led the ingress modernization decision for the AKS platform, replacing the NGINX Ingress Controller with Azure Application Gateway Ingress Controller.** Evaluated Application Gateway for Containers against AGIC across capability maturity, compatibility, security, scalability, and operational fit, documented the trade-offs, and selected AGIC as the production architecture for the current landscape.
- **Architected an Azure AI Search–based retrieval platform for Microsoft marketing websites, replacing 3rd party ElasticPress.** Designed the end-to-end content ingestion, chunking, embedding, indexing, and retrieval architecture, combining lexical, vector, and semantic ranking to improve relevance and content discoverability while eliminating the operational overhead of self-managed search infrastructure.
- **Established a reusable Azure DevOps platform engineering capability** with standardized pipeline templates, environment promotion patterns, automated quality gates, CodeQL SAST, Microsoft Defender controls, and test automation. Codified infrastructure through Terraform to improve deployment consistency, reduce release risk, and accelerate delivery across environments.
- **Embedded security, privacy, and resilience into application design and delivery** by partnering with project development teams, Microsoft Security, and third-party vendors to create and review threat models, complete privacy reviews and SDL assessments, establish BCDR readiness, and align external delivery teams with Microsoft security standards and controls for a consistent, secure, and efficient SDLC; implemented continuous security validation through SAST with CodeQL, dependency and supply-chain analysis with Dependabot and Microsoft Defender, and DAST using Microsoft proprietary tools including WCP and URSA, with vulnerability triage and remediation aligned to MSRC processes.
- **Provided operational governance and leadership across incident, service request, security, and release management**, establishing clear ownership, prioritization, escalation, and execution practices; regularly monitored Microsoft S360 for security findings, coordinated remediation with engineering teams, and drove issues through timely closure.

- **Automated day-to-day SRE operations** with custom tooling built on Azure Automation, Python, LangChain, and Azure OpenAI, and built centralized monitoring dashboards with Datadog, Prometheus, Grafana, and Azure Monitor.

Principal Azure AI Solution Architect — GenAI Enablement Practice

Feb 2026 – Present

Founding Azure AI solution architect for a GenAI enablement practice within Microsoft's Business Unit, defining the reference architecture, governance model, engineering standards, and adoption roadmap required to move generative AI use cases from experimentation into secure, observable, production-grade solutions across development, support, and DevOps organizations.

- **Defined the practice's Azure AI reference architecture, technology selection criteria, responsible-AI controls, and phased adoption roadmap.** Partnered with product and engineering leaders to assess use-case value, data readiness, security, integration complexity, and operating-model impact, prioritizing solutions with measurable productivity and service-management outcomes.
- **Architected and delivered a portfolio of production AI solutions using agentic and retrieval-augmented generation patterns.** SRE Buddy integrates operational SOPs with ICM, Kusto, and Azure DevOps to accelerate incident investigation; AICA analyzes comments from Microsoft blog sites and marketing chatbots to convert customer signals into management insights; and Microsoft NLWeb enables conversational, natural-language discovery across marketing web properties to improve content access and engagement.
- **Designed and implemented an Azure SRE Agent architecture spanning ticket classification, resolver-group routing, and first-line diagnostics,** including log retrieval, health validation, and known-error matching before engineer engagement. Reduced manual triage effort and established a phased roadmap for governed runbook execution and automated resolution of recurring incident classes to further reduce MTTR.
- **Defined the solution approach for LLM fine-tuning to automate incident cause-code classification,** including the target taxonomy, training-data preparation, evaluation criteria, and production-operating considerations. The solution is projected to eliminate one full-time equivalent of manual classification effort while improving categorization consistency and accuracy.
- **Architected an AKS investigation copilot using the Grafana observability stack and a Grafana MCP-powered conversational interface.** Enabled engineers to query cluster health, telemetry, and dashboards in natural language, translating operational intent into governed observability actions and shortening diagnosis time during SRE triage.
- **Built a consolidated status-reporting dashboard** giving leadership a single-pane view of incidents, DevOps user stories, and S360 KPIs via Microsoft MCP servers — streamlining executive status reporting and team tracking, extending it with Grafana and Azure MCP servers to fold infrastructure and application monitoring into the same view.
- **Established a reusable AKS-based hosting and CI/CD architecture for GenAI workloads,** with Langfuse and Grafana providing prompt and response tracing, evaluation, operational telemetry, governance, and guardrail monitoring across the AI solution lifecycle.
- **Drove GenAI enablement and responsible-use standards,** coordinating training with vendors and SMEs to upskill multiple engineering teams.

Tenet Healthcare (via NTT Data / Dell)

Dallas, TX

Technical Architect & Delivery Lead — DevOps & App Modernization

Nov 2021 – Sep 2022

Modernized and standardized Tenet's application delivery platform through enterprise-wide Azure DevOps adoption — migrating legacy TFS, strengthening security and compliance, and establishing scalable CI/CD across diverse technology stacks to improve developer productivity, release reliability, and visibility.

- **Spearheaded enterprise Azure DevOps adoption,** migrating 60+ projects from legacy TFS — streamlining version control (Git), work-item tracking (Azure Boards), and reporting (Azure Monitor, Power BI) across the account.
- **Developed and optimized YAML-based CI/CD pipelines** for .NET, Node.js, Java, and Python applications, integrating AKS, Helm, and Docker containerization; instituted a Git-flow branching strategy with cross-functional teams to modernize the development process.
- **Designed and enforced security controls** — Azure AD, RBAC, service connections, pipeline approvals, and branch policies — and integrated SonarQube code scanning into pipelines to enforce Secure SDLC compliance.
- **Led migration of 48 consumer websites to a custom ASP.NET Core / Vue.js / Sitefinity CMS,** raising availability to 99.99% and Lighthouse scores from single digits to 80%.
- **Ensured a seamless, low-risk transition** through hands-on technical guidance and troubleshooting, and conducted training and workshops on Azure DevOps best practices, security automation, and cloud-native deployment strategies.

Tower Leader — Enterprise App Support & SharePoint Modernization

May 2015 – Nov 2021

Owned a \$5M/year enterprise portfolio (consumer sites, SharePoint portals, BPM apps) serving 160K users across 48 hospitals; led onshore/offshore teams of 20+.

- **Directed incident and problem management under ITIL,** reducing ticket volume 60% over two years through automation and root-cause analysis.
- **Managed customer feedback, escalations, and executive stakeholder communications,** ensuring SLA compliance and alignment with business priorities.
- **Transitioned applications from third-party vendors to in-house support,** reducing operational cost and improving service continuity.
- **Led consolidation and migration of a 4TB+ SharePoint estate** (40+ hospital intranets, 40+ consumer sites) with governance, access control, and end-user training.

Dell Services

Bengaluru, India

Practice Delivery Leader — Microsoft SharePoint Practice

Apr 2013 – Apr 2015

Built and led Dell's SharePoint practice within the Microsoft horizontal, managing a 25-person team across Bengaluru and Noida.

- **Institutionalized delivery excellence** — reusable asset repository, coding standards, governance, and operational reviews.

- **Developed fixed-price service offerings** (SharePoint migration, knowledge management, ITIL) with sales collateral, supporting pre-sales and RFP/RFI response.
- **Led capability building** through training, certification, and demand-supply planning; partnered with Dell Software Group on optimized delivery models.

Project Lead — BizTalk Orchestration (Owens & Minor)

Nov 2011 – Mar 2013

- **Led offshore delivery of a centralized BizTalk ESB layer** replacing point-to-point integrations; owned WBS, estimates, quality reviews, audits, and stakeholder reporting.

Team Lead — Microsoft Amalga EMR Program

Sep 2009 – Nov 2011

- **Built Amalga UIS competency for Dell and led implementations** at Moses Cone, Norton Healthcare, and St. Joe Hospitals using HL7/XML standards, managed offshore delivery and quality.

EDUCATION & CERTIFICATIONS

- **Post Graduate Diploma in IT (Executive Management)**, VGSOM, IIT Kharagpur, India
- **B. Tech, Electrical Engineering**, CET, OUAT Bhubaneswar, India
- **Postgraduate in AI/ML & GenAI** — In Progress

Certifications: AZ-400 (DevOps Engineer Expert) AZ-104 (Azure Administrator) CKAD · Terraform Associate · PMP · PSM I (Scrum Master) HL7 Healthcare Messaging